Jan. 31, 1967 A. M. GILBERT 3,301,119

DRUMSTICK

Filed Feb. 10, 1966

FIG. 1

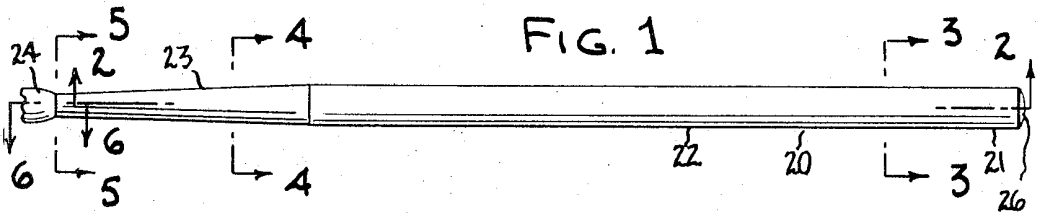


FIG. 2

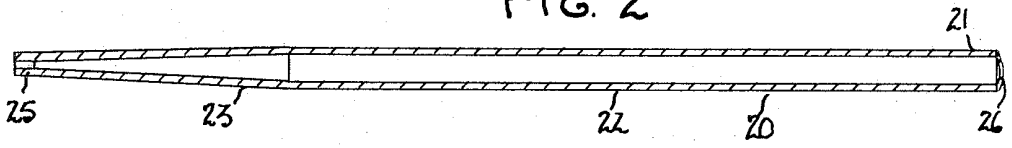


FIG. 9

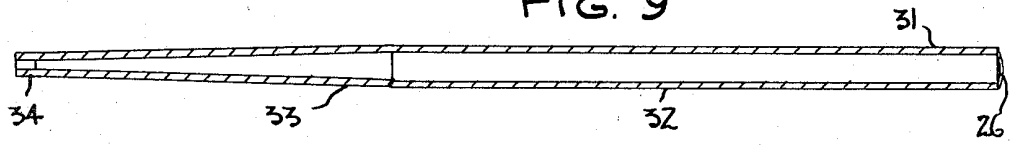


FIG. 10

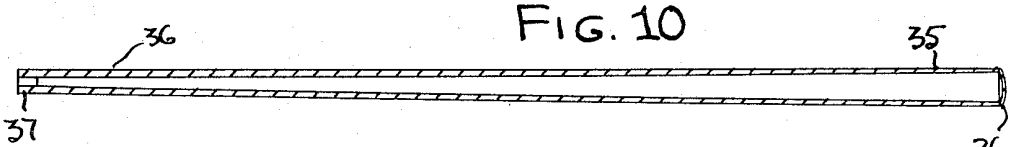


FIG. 3

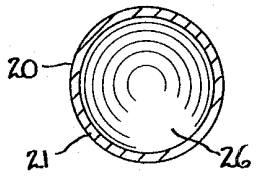


FIG. 4

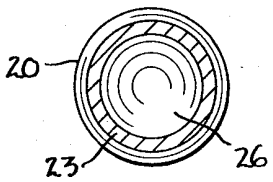


FIG. 5

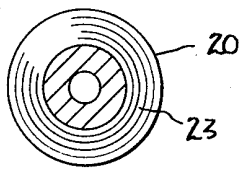
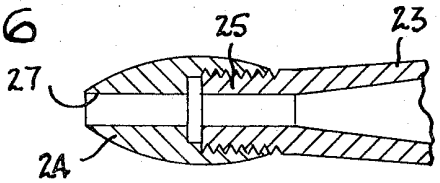


FIG. 6



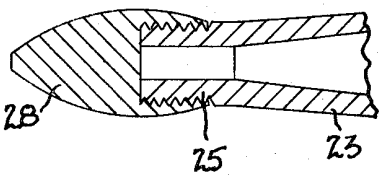
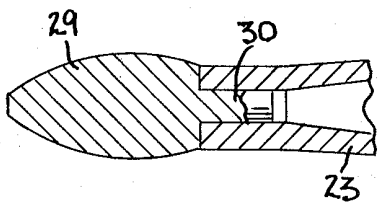


FIG. 7

FIG. 8



INVENTOR.
ANNA MARIE GILBERT
BY
Isler & Dinstein
ATTORNEYS.

United States Patent Office 3,301,119
Patented Jan. 31, 1967

3,301,119

DRUMSTICK

Anna Marie Gilbert, University Heights, Ohio
(366 Lourdes Lane, Costa Mesa, Calif. 92627)

Filed Feb. 10, 1966, Ser. No. 526,553
9 Claims. (Cl. 84—422)

The present invention relates generally to drumsticks used in association with or as part of percussion instruments such as drums, cymbals, wood blocks and the like. More particularly, the present invention relates to a hollow, balanced drumstick of uniform weight distribution which is designed to provide greater durability, greater maneuverability and superior performance over known forms of drumsticks of the prior art.

For a great number of years, the conventional wooden drumstick was the best known and most accepted drumstick in the percussion field. Conventionally, these wooden drumsticks range from approximately 15 inches to approximately 18 inches in length and have a butt and shaft portion having a nominal diameter of ½ inch. The butt and shaft portions represent approximately 85% of the total length of the drumstick, the remaining portions being a tapered or conical shoulder portion which terminates in an enlarged bead portion, which provides the conventional impact point or element of the drumstick. It is considered very desirable that each of a pair of drumsticks be matched for tone, weight and balance, and that the drumsticks be linearly straight or true.

There are several factors which determine the usefulness and percussion efficiency of a drumstick. Generally, the ideal drumstick would have the following attributes:

(1) It would have durability and longevity. The drumstick should be capable of being used for the purposes for which it is intended without the danger of splitting, chipping or breakage, particularly at the narrowed portion where the bead connects to the shoulder.

(2) The drumstick should have stability in the sense that its characteristics should remain uniform indefinitely, so that (a) a pair of drumsticks can be matched perfectly and (b) the matching characteristics will remain stable and will not deteriorate.

(3) The drumsticks should have a high degree of maneuverability and response, so that the fastest possible contact with the drumhead can be attained, as well as a resilient and rapid rebound from the drumhead.

(4) The drumstick should not distort or dull the tone produced by the drum. The drumstick should have pinpoint definition and a maximum of percussion dynamics, so as to provide sharpness and brilliance in its contact with the drumhead and thus obtain the true drum sound and vibration.

The conventional wooden drumstick was evolved over a period of many years and achieved its present design so as to produce excellent results with respect to the latter two factors mentioned above, although any further improvement in these characteristics would be of great benefit to the drummer. An increase in maneuverability would be such an improvement and could be achieved by improving the balance of conventional drumsticks. However, it is not possible to achieve a perfectly balanced wooden drumstick due to the physical characteristics predetermined by its design. The attempts made in the prior art to alter the balance resulted in creating concentrated areas of imbalance, detrimental to maneuverability, response and tonal quality. Moreover, the wooden drumstick is greatly deficient and inadequate with respect to factors number (1) and (2) mentioned above. It is difficult to match a pair of drumsticks, due to differences in the wood, and even when a pair is properly matched, this matching is of an impermanent nature because the drumsticks are affected by use and weather, which creates chipping and warping. Furthermore, the wooden drumsticks have not overcome the problem of splitting and breakage, particularly the fracturing of the drumstick at the narrowed portion where the bead connects to the shoulder.

Although there have been attempts in the prior art to overcome these latter disadvantages of the wooden drumsticks, by making drumsticks out of solid metal, or by making them out of combinations of hollow metal and wood or synthetic resin, or by making them out of metal tubes with weighted portions, these attempts have not been entirely successful. They have overcome the problems of durability, but in doing so have sacrificed one or more of the characteristics in which the wooden drumstick still excels.

It is a primary object of my invention to provide a hollow, balanced drumstick of uniform weight distribution which will achieve the weight and feel of the wooden drumstick while achieving characteristics of durability, stability and maneuverability superior to those of wooden drumsticks.

Another object of my invention is to provide a drumstick of the character described, which will be economical to manufacture.

Still another object of my invention is to provide a drumstick of the character described with replaceable beads.

Other objects and advantages of my invention will become apparent during the course of the following description.

In the drawings, in which like reference numerals designate like parts throughout the same,

FIG. 1 is a plan view of a drumstick embodying the features of my invention.

FIG. 2 is a lonigtudinal cross-sectional view taken as indicated on line **2**—**2** of FIG. 1.

FIG. 3 is an enlarged transverse cross-sectional view, taken as indicated on line **3**—**3** of FIG. 1.

FIG. 4 is an enlarged transverse cross-sectional view, taken as indicated on line **4**—**4** of FIG. 1.

FIG. 5 is an enlarged transverse cross-sectional view, taken as indicated on line **5**—**5** of FIG. 1.

FIG. 6 is an enlarged fragmentary cross-sectional view, taken as indicated on line **6**—**6** of FIG. 1.

FIG. 7 is a view similar to FIG. 6, but showing a modified form of bead.

FIG. 8 is a view similar to FIG. 6, but showing still another form of bead.

FIG. 9 is a longitudinal cross-sectional view similar to that of FIG. 2, but showing a modified configuration of the drumstick.

FIG. 10 is a longitudinal cross-sectional view similar to that of FIG. 2, but showing still another modified form of the drumstick.

Referring more particularly to FIGS. 1 through 5 of drawings, I have shown a hollow, tubular drumstick **20**, having a butt portion **21**, a shaft portion **22**, a shoulder portion **23**, and a bead portion **24**.

Both the butt **21** and shaft **22** are cylindrical in form and are of uniform diameter, both externally and internally. By way of example, the outside diameter of this cylindrical portion may be .500 inch and the inside diameter may be .4375 inch, thus providing a wall thickness of .03125 inch. This cylindrical portion consisting of the butt **21** and shaft **22** may have a length of 11 inches and will terminate at the conical shoulder portion **23**. The shoulder portion has a tapered or conical configuration which ranges from an outside diameter of .500 inch at its point of juncture with the shaft **22** to .250 inch at the terminus of the taper. The internal diameter of the shoulder portion ranges from .4375 inch at its juncture with the shaft **22** to .0625 inch at the terminus of the taper. Thus, the wall thickness of the shoulder

portion will range from .03125 inch at its juncture with the shaft 22 to .09375 inch at the terminus of the tapered portion. It is important to note that this graduated increase in wall thickness in the shoulder portion 23 is predetermined to provide a uniform solid cross-sectional area of .04602 inch in any diametrical section of the shoulder portion 23. This same area of solid diametrical section also exists in the butt portion 21 and shaft portion 22 of the drumstick. Therefore, any unit incremental length of the body of the drumstick has exactly the same mass or weight. By progressively increasing the wall thickness of the tapered portion of the drumstick as the diameter thereof decreases, uniformity of unit weight is achieved and precise balance of the drumstick is maintained.

warpage and has its greatest wall thickness at its bead end, which is the impact end heretofore most vulnerable to fracture or splintering. In contrast to other prior art forms, the drumstick of my invention has no localized concentrations of weight which provide a concentrated area of imbalance and the body of the drumstick is a homogeneous unit.

I have conducted balance tests on four different models of wooden drumsticks, as well as on two different sizes of commercially sold composite drumsticks made in accordance with the teachings of prior art Patent No. 3,146,659 to Robba et al. The results of these balance tests have been compared to the results of a like balance test on a drumstick made in accordance with my invention, and the results are listed below:

Type of drumstick	Total Length (inches)	Distance Between Butt end and point of balance (inches)	Distance Between Bead end and point of balance (inches)	Percentage of length between Butt end and point of balance	Percentage of length between bead end and point of balance
Wooden Stick, Type 1	15.936	6.968	8.968	43.5	56.5
Wooden Stick, Type 2	16.218	7.468	8.750	42.0	58.0
Wooden Stick, Type 3	16.375	7.312	9.062	44.0	56.0
Wooden Stick, Type 4	15.843	7.031	8.812	44.0	56.0
Verisonic (Robba), ½" O. D	15.250	7.937	7.312	57.3	42.7
Verisonic (Robba), ¾" O. D	15.531	8.125	7.406	57.3	42.7
Gilbert, ½" O. D	15.656	7.781	7.875	49.7	50.3

The shoulder portion 23 is provided with a cylindrical extension 25 whose inside and outside diameters are the same as the smallest diameters of the tapered portion of the shoulder. The extension 25 provides a bead-supporting portion, which is externally threaded to receive the internally threaded bead 24. The shoulder may have a length of 4.25 inches, including the extension 25, which may be approximately .25 inch in length. When the bead 24 is threadedly secured to the extension 25, the bead will project .375 inch beyond the end of the extension, thus providing a drumstick which is 15.625 inches in overall length. It will be understood that the bead may be somewhat longer or somewhat shorter than in the example given.

The open end of the butt 21 may be plugged or closed, if desired, by a disk 26 of any suitable lightweight material which will not significantly affect the weight distribution which has been herein described.

Referring to FIG. 6 of the drawings, it will be noted that the bead 24 is provided with a bore or hollowed out portion 27, which can be made of such size and configuration as will effectively reduce the weight or mass of the bead 24 to a value which will maintain the weight uniformity and balance of the drumstick, as previously described.

I have found that the use of aluminum alloy tubing as the material for the drumstick, is preferable, as it is sufficiently light in weight, durable and malleable for this purpose. If the largest outside diameter of the drumstick body is to be .500 inch, a piece of aluminum alloy tubing of that outside diameter, and of appropriate length, and having a wall thickness on the order of .0350 inch, can be subjected to a rotary swaging process to obtain the necessary reduction in outside diameter and the accompanying increase in wall thickness to achieve the uniform weight distribution that I have described. When the above described drumstick body is formed of 2024T4 aluminum alloy, each one inch increment of its length will weigh .005 pound throughout its length of 15¼ inches for a total weight of .07625 pound for the body itself. The bead may add an additional .003 pound and the plug 26 on the opposite end of the body may add the same amount of weight for a total weight of .08225 pound. This total weight is distributed evenly throughout the entire length of the drumstick and thus provides the balanced drumstick which, in contrast to the conventional wooden drumstick, is durable and not subject to

Refering now to FIG. 7 of the drawings, I have shown a fragmentary cross-sectional view of another form of bead 28, which is adapted for threaded securement to the extension 25 of the drumstick body. In contrast to the bead 24, previously described, the bead 28 is somewhat shorter in length and is of solid cross-section, rather than being hollowed out for weight reduction.

In FIG. 8 of the drawings, I have shown another form of bead 29, which is provided with an extension 30 of reduced diameter, which is adapted to be received in the bore of the extension 25 by means of a press fit. The type of bead shown in FIG. 8 would generally be considered as permanently affixed to the drumstick. The type of bead illustrated in FIGS. 6 and 7 of the drawings would generally be considered as replaceable and would permit the drummer to change the size or configuration of the bead on the drumstick to suit varying progression requirements.

In FIG. 9 of the drawings I have shown a modified form of body for the drumstick, which has an overall length of 15¼ inches, the same as the drumstick body illustrated in FIG. 1 of the drawings, except that the proportion of the shoulder length to the shaft and butt length has been changed. In the form shown in FIG. 9, the butt portion 31 and the shaft portion 32, which are of cylindrical configuration, have a combined length of 5½ inches. The tapered shoulder portion 33 has a length of 9½ inches and terminates in the cylindrical bead-supporting extension 34 which is ¼ inch in length. In this form of the invention the angle of taper on the shoulder portion 33 is less acute than the angle of taper on the shoulder portion 23 in the form shown in FIG. 1. However, the form of FIG. 9 embodies the same principles of invention, in the sense that the wall thickness of the body of the drumstick is graduated to an increased thickness as the outside diameter of the shoulder portion is reduced, so that each unit section of the length of the drumstick body is of identical weight to any other unit section.

FIG. 10 shows still another modified form of the drumstick body, in which the entire body is tapered from the butt end 35 to the shoulder end 36, which as heretofore, terminates in the cylindrical bead-supporting extension 37. As in the forms illustrated in FIG. 1 and in FIG. 9, the tapered drumstick body of FIG. 10 has a uniform weight distribution in which each unit length of the body

is of identical weight to any other unit length thereof.

Although, in describing the features of a drumstick embodying my invention, I have given illustrative examples describing particular dimensions, weights and materials used, as well as various forms of tapers, it will be understood that such examples are by way of illustration only and that the principles of my invention can be utilized with materials other than those I have described and with various dimensional characteristics as to length, diameter, wall thickness and the like.

It is to be understood that the forms of my invention, herewith shown and described, are to be taken as preferred examples of the same, and that various changes in the shape, size and arrangement of parts may be resorted to, without departing from the spirit of my invention or the scope of the subjoined claims.

Having thus described my invention, I claim:

1. A hollow unitary drumstick body of tubular form having a butt portion and a tapered shoulder portion, the outside diameter of said butt portion being greater than the smallest outside diameter of said shoulder portion, and the wall thickness of said body being graduated to increasing values in proportion to decreasing values of said outside diameter to maintain a substantially uniform diametrical cross-sectional area of wall throughout the length of said body.

2. A hollow unitary drumstick body as defined in claim 1, wherein said shoulder portion is of conical configuration and is of a length which is less than 40% of the total length of said body.

3. A hollow unitary drumstick body as defined in claim 1, wherein said butt portion is of conical configuration.

4. A hollow unitary drumstick body as defined in claim 1, wherein each incremental unit length thereof is of equal weight.

5. A hollow unitary drumstick body as defined in claim 4, wherein aluminum alloy tubing is utilized as the material thereof.

6. A hollow drumstick of tubular form having a unitary body, said body having a butt portion and a tapered shoulder portion, the outside diameter of said butt portion being greater than the smallest outside diameter of said shoulder portion, the wall thickness of said body being graduated to increasing values in proportion to decreasing values of said outside diameter thereof to maintain a substantially uniform diametrical cross-sectional area of wall throughout the length of said body, and a bead carried by the terminal end of said shoulder portion.

7. A hollow drumstick as defined in claim 6, wherein said bead is fixedly secured to said body.

8. A hollow drumstick as defined in claim 6, wherein said bead is provided with a weight-reducing bore extending therethrough.

9. A hollow drumstick as defined in claim 6, wherein said bead is removably secured to said body.

References Cited by the Examiner

UNITED STATES PATENTS

2,521,336	9/1950	Bramson	84—422
3,146,659	9/1964	Robba et al.	84—422

RICHARD B. WILKINSON, *Primary Examiner.*